SpaceX Compliance: licenses, spectrum, insurance, and non-English primary law

Search Audit · 100 prompts · 6 providers · run 2026-10-09

Parallel gives SpaceX the best balance of accuracy, cost and speed. 4 providers tie on accuracy, so cost and speed decide.

MOST CORRECT

Parallel: 61 of 100

Tied within chance with Exa, Grok native and Keenable router.

CHEAPEST OF THE MOST CORRECT

Parallel: $0.440 per prompt

61 of 100 correct.

FASTEST

Exa: 43.9s median

60 of 100 correct.

Ranking

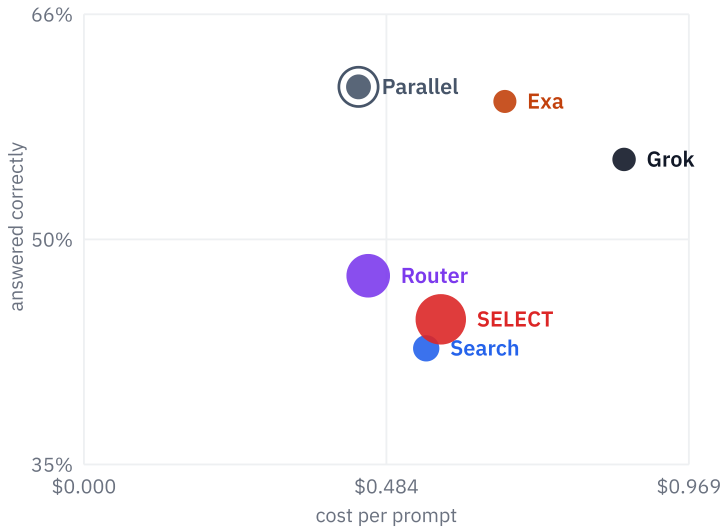
Value score combines accuracy, cost and speed per prompt (wrong = 0). Weights: accuracy 60, cost 25, speed 15.

#	Provider	Value score	Correct	$ / prompt	Time
1	Parallel	65	61/100 tied	$0.440	56.7s p90 5.1m
2	Exa	63	60/100 tied	$0.674	43.9s p90 5.0m
3	Grok native	57	56/100 tied	$0.865	44.5s p90 4.8m
4	Keenable router	55	48/100 tied	~$0.455	3.6m p90 12.4m
5	Keenable search	53	43/100	~$0.548	1.2m p90 9.0m
6	Keenable SELECT	50	45/100	~$0.571	4.6m p90 15.5m

green marks the best on this table. A tie means the accuracy gap is within chance (paired McNemar test, Holm-corrected, vs Parallel). ~ = priced from a list price.

Accuracy vs cost

Up and left is better. Dot size = median time. Ringed = Pareto front: no other provider is both at least as accurate and at least as cheap.



Cost & sources

Cost break-even vs Exa

A cheaper price per request only wins if the agent doesn't need proportionally more requests. A provider is cheaper on search than Exa when its **request ratio** (its requests per prompt ÷ Exa's) is below its **price advantage** (Exa's price per request ÷ its own). Requests = searches + page fetches (+ SQL queries for SELECT). Model tokens are excluded here; they are in the ranking.

Provider	Price / 1k requests	Requests / prompt	Price advantage	Request ratio	Search $ / prompt	Search $ / correct answer
Exa	$2.29	20.3	1.00×	1.00×	$0.0463	$0.0772
Parallel	$1.99	15.7	1.15×	0.78×	$0.0314	$0.0515
Keenable search	~$3.88	33.1	0.59×	1.63×	~$0.1283	~$0.2984
Keenable SELECT	~$3.57	24.1	0.64×	1.19×	~$0.0858	~$0.1908
Grok native	$3.02	15.6	0.76×	0.77×	$0.0472	$0.0843
Keenable router	~$3.70	23.1	0.62×	1.14×	~$0.0855	~$0.1781

By prompt type: request ratio vs price advantage

Green = cheaper on search than Exa for that type (request ratio below price advantage).

Type	Parallel	Search	SELECT	Grok	Router
Multi-hop · 23	0.75× / 1.12×	2.08× / 0.85×	1.58× / 0.96×	1.10× / 1.07×	1.48× / 0.89×
List-building · 31	0.87× / 1.14×	1.53× / 0.39×	1.20× / 0.43×	0.71× / 0.68×	1.14× / 0.42×
Regulation · 46	0.60× / 1.00×	1.57× / 0.83×	0.91× / 0.87×	0.69× / 0.70×	0.93× / 0.85×

Sources each provider's answers cite

Domains cited in final answers. "Only here" = domains no other provider cited anywhere in this audit, a rough measure of index coverage. "Wrong, uncited" = wrong answers with no source at all (often answered from memory after search failed).

Provider	Sources / answer	Distinct domains	Only here	Uncited answers	Wrong, uncited	Most cited
Exa	21.0	467	151	0/100	0/16	docs.fcc.gov 51 fcc.gov 39 fcc.report 35 govinfo.gov 32 federalregister.gov 28 faa.gov 23
Parallel	22.3	404	191	0/100	0/12	docs.fcc.gov 56 fcc.gov 43 facebook.com 35 govinfo.gov 30 federalregister.gov 26 satellitetoday.com 21
Keenable search	4.0	122	9	2/100	2/18	docs.fcc.gov 49 fcc.gov 12 ecfr.gov 11 faa.gov 9 govinfo.gov 9 federalregister.gov 7
Keenable SELECT	3.8	116	10	1/100	1/18	docs.fcc.gov 47 ecfr.gov 10 govinfo.gov 10 fcc.gov 9 faa.gov 9 fcc.report 7
Grok native	20.8	434	118	0/100	0/19	docs.fcc.gov 48 fcc.gov 41 govinfo.gov 31 fcc.report 29 federalregister.gov 26 faa.gov 23

Provider	Sources / answer	Distinct domains	Only here	Uncited answers	Wrong, uncited	Most cited
Keenable router	3.7	102	0	2/100	0/13	docs.fcc.gov 48 govinfo.gov 12 fcc.gov 10 ecfr.gov 9 faa.gov 9 law.cornell.edu 6

How we measured

Setup

Prompts	100 · run 2026-10-09
Agent model	`x-ai/grok-4.7` pinned in the audit config
Judge model	`anthropic/claude-sonnet-5.5`
Budget per answer	max tool calls 100, max seconds 1800
Cost of this audit	$366.61

Providers

Exa	OpenRouter openrouter:web_search + openrouter:web_fetch, engine=exa $0.007 per search, $0.001 per fetched page (OpenRouter list price for Exa)
Parallel	OpenRouter openrouter:web_search + openrouter:web_fetch, engine=parallel $0.005 per search, $0.001 per fetched page (OpenRouter list price for Parallel)
Keenable search	Keenable REST /v1/search + /v1/fetch as function tools $0.004 per request (Keenable list price)
Keenable SELECT	Keenable SELECT MCP (https://select.keenable.ai/mcp); the agent writes the SQL the tool describes, plus Keenable fetch $0.004 per request (Keenable list price)
Grok native	OpenRouter openrouter:web_search + openrouter:web_fetch, engine=native Passed through inside the OpenRouter completion charge. No separate list price for native search.
Keenable router	Jev answers one choice, search or select. The agent then has only that tool $0.004 per request (Keenable list price)

CAVEATS

- Every provider ran the same prompts with the same pinned agent model, system prompt and temperature (0.2).
- Budget per run: 100 tool calls and 1800 seconds; runs that hit it were forced to answer and are flagged.
- Exa, Parallel and Grok native ran as OpenRouter server tools (openrouter:web_search and web_fetch), so OpenRouter's own loop decides how often they search inside one model call. Keenable ran as function tools in our own agent loop. The server-tool providers' search cost is what OpenRouter billed, split from the model cost when OpenRouter reported the split, otherwise estimated from list prices (marked ~).
- Keenable search and SELECT are priced at the list price of $0.004 per request. Requests that failed are priced at $0, so an effective rate can sit below list.
- Keenable SELECT is experimental: the agent is given a research skill and writes DuckDB SQL (WEB_SEARCH, SEM_EXTRACT) in the query argument. The other providers do not get that skill. A failed query still counts toward the tool-call budget and is priced at $0, since it is not known whether Keenable bills it.
- Keenable router asks Jev (typesafe/jev-1.13, a small decision model served through OpenRouter) one choice question, search or select, and returns the answer that arm already gave, plus the cost and time of the Jev call. A prompt where either arm failed has no router run and counts as wrong.
- Prompt ids run past 100 with gaps: on 2026-10-09 the 45 prompts that every provider had answered correctly were removed, since they could not separate anyone, and the same number of harder cross-document prompts were added. Ids are never reused.
- Exa: 4 returned no answer, of 100 runs; Grok native: 6 returned no answer, of 100 runs; Keenable search: 1 run errored and 11 hit the budget, of 100 runs; Keenable router: 5 hit the budget, of 98 runs; Keenable SELECT: 1 run errored and 7 hit the budget, of 100 runs. A run that errored or returned no answer counts as wrong; a run that hit the budget was made to answer and graded like any other.
- Judge: anthropic/claude-sonnet-5.5, bundle judge-v3, temperature 0, sees the reference answer.
- One run per prompt is the scored trial. Rerun the audit before treating the ranking as current: providers, models and the web change. About 100 prompts is the smallest set with a real chance of detecting a 10-point accuracy gap. A tie on fewer prompts means the sample was too small to see that gap, not that the providers are equal. The set is 100 prompts because each provider run costs real money; a gap of a few points here is a lead to test at a larger size, not a verdict.
- The judge was not calibrated against hand-graded answers in this audit, so treat small accuracy gaps between providers with care.